

- Introduction to Probability
Lec 4-6 Problem Sets

- 6.041 Probabilistic Systems Analysis - Problem Set 3.

1) n persons; pick hats at random; equally likely.
(no replacement)

a) total outcomes = ~~$n!$~~ $n!$
specific event (each gets own hat) = 1 {equally likely}
 $\therefore P(\text{own hat}) = \frac{1}{n!}$ ✓

b) First m get own hat.

$$\text{or } \frac{n!}{(n-m)!} = \frac{1 \cdot (n-m)!}{n!}$$

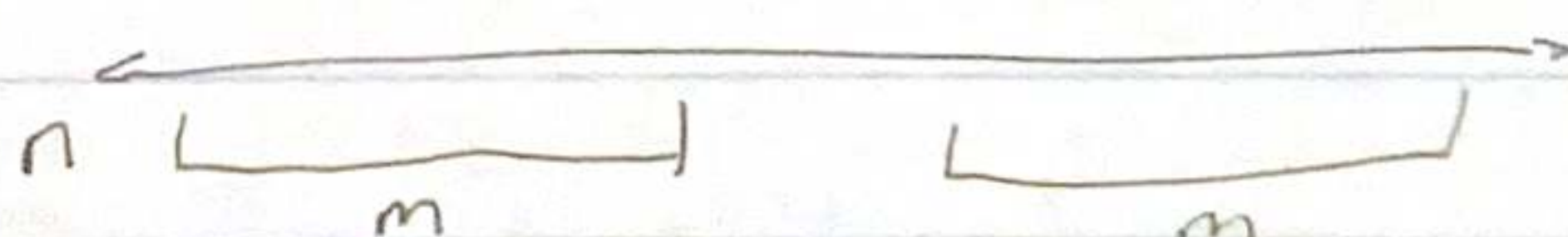
$$P(\text{1st } m \text{ own hat}) = \frac{\left[\frac{n!}{(n-m)!} \right]}{n!} \quad \text{total} = \frac{1}{n!} \quad (6-3)!$$

$$\text{desired} = \frac{(n-m)!}{n!}$$

$$P = \frac{(n-m)!}{n!}$$

~~$\left[\frac{n!}{(n-m)!} \right] / \left[\frac{1}{n!} \right]$~~

c) $n! : (n-m)! m!$
 $\Rightarrow \frac{m! (n-m)!}{n!}$
 $\Rightarrow \frac{m! (n-m)!}{n!}$
 $\therefore P = \frac{m! (n-m)!}{n!}$ ✓



d) p = gets dirty ~~p~~ $(1-p)m!$ ~~$(n-m)!$~~
 first m clean hats $\Rightarrow (1-p)m!$ ✗
 $\rightarrow P(\text{clean}) = (1-p)$
 $\hookrightarrow$ independent
 $P(m \text{ clean}) = (1-p)(1-p) \dots (1-p) = \underline{(1-p)^m}$

e) $P(\text{exactly } m \text{ clean}) = ?$

$m \text{ clean} \Rightarrow (1-p)^m$

$n-m \text{ dirty} \Rightarrow p^{n-m}$

$\therefore P(\text{exactly } m \text{ clean}) = (1-p)^m p^{n-m} \times$
 $= (1-p)^m p^{n-m} \binom{n}{m}$

↳ Key insight: no longer first m as in d).

↳ So order does not matter + no repetition → Combination

↳ $\binom{n}{m}$

Each prob of event is: $(1-p)^m p^{n-m}$
 # events is: $\binom{n}{m}$

↳ In (d) it was a specific m people $\Rightarrow$ so 1 event.



2.) A → Randomly choose 4 cards; 52 card deck.

B → R. choose 8 cards

$\frac{8!}{52} \cdot \frac{8!}{51} \cdot \frac{1}{50} \cdot \frac{1}{49} \cdot \text{rest.}$

$P(\text{B choose all A's cards}) =$

~~$\frac{8!}{52} \cdot \frac{8!}{51} \cdot \frac{1}{50} \cdot \frac{1}{49} \cdot \text{rest.}$~~ $\frac{8}{52}$

$\frac{52!}{8!4!} \cdot \frac{1}{\binom{52}{8} \binom{52}{4}} = P = \frac{\binom{8}{4} \binom{52}{4}}{\binom{52}{8}}$

4 cards of 8 are fixed

↳ so select 4 additional from 48

$P = \frac{\binom{4}{4} \binom{48}{4}}{\binom{52}{8}}$

Introduction to Probability Textbook Problem Sets

Lecture 4-6 Problems

Problem Set - Section 1.6 - Counting

49) Six-sided die; Roll 3x times independently;
Which is more likely: Sum = 11 or Sum = 12?
{1, 2, 3, 4, 5, 6}

11:

1. 2+4+5 (2) (1-1)
2. 1+5+5 (1) (1-2)
3. 3+3+5 (3) (1-3)
4. 4+1+6 (1) (1-4)
5. 2+3+6 (3) (1-5)
- ~~6. 3+2+6 (2) (1-6)~~
6. 4+4+3 (1) (1-6)

12:

1. ~~4+4+4~~ (4) (12-1)
2. ~~3+4+5~~ (3) (12-2)
3. 1+5+6 (1) (12-3)
4. 2+5+5 (2) (12-4)
5. 2+4+6 (2) (12-5)
6. 3+3+6 (3) (12-6)

lowest: 3
highest: 18

#11:

~~#11: Count(11-1) = $\frac{n}{1} = \frac{18}{3} = 6$~~
~~2+4+5~~
~~1+5+5~~
~~3+3+5~~
~~4+1+6~~
~~2+3+6~~
~~3+2+6~~
~~4+4+3~~

Count(11-1) = 3!

#11: $6 \cdot 3! = 36$
 #12: $6 \cdot 3! = 36$

11-1: $3! = 6$	12-1: $1 = 1$
11-2: $\frac{3!}{2!} = 3$	12-2: $3! = 6$
11-3: $3! = 6$	12-3: $3! = 6$
11-4: $3! = 6$	12-4: $3 = 3$
11-5: $3! = 6$	12-5: $3! = 6$
11-6: $3! = 6$	12-6: $3 = 3$
<u>27</u>	<u>25</u>

total outcomes = $6 \cdot 6 \cdot 6 = 6^3 = 216$

$P(\text{Sum} = 11) = \frac{27}{216}$

$P(\text{Sum} = 12) = \frac{25}{216}$

∴ $P(\text{Sum} = 11) > P(\text{Sum} = 12)$

Sum = 11 more likely ✓

50) n people at party; equal prob any born same day of year. independently.

$$P(\text{distinct birthday everyone}) = ?$$

total outcomes = $\binom{365}{n}$ Days (365) $\rightarrow \rightarrow \rightarrow \rightarrow \rightarrow$

$$P(\text{distinct}) = \frac{\binom{365}{n}}{365^n}$$

total outcomes = 365^n ✓

(assume $n \leq 365$)

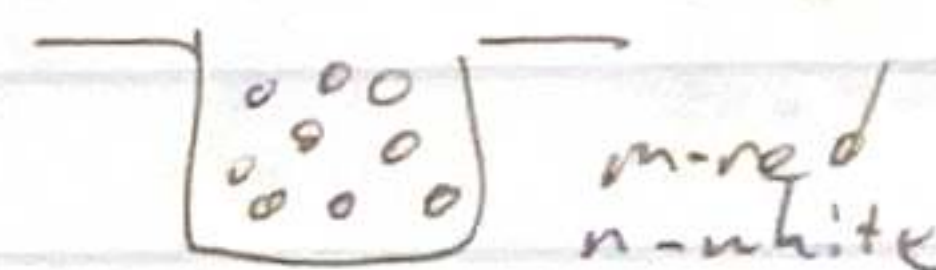
$$\frac{365!}{(365-n)!}$$

51) Urn contains m red and n white balls

a) Draw two balls at once.

$R+R$; $R+W$; $W+W$

total balls = $m+n$



total outcomes = $\binom{m+n}{2}$ X

total outcomes = $(m+n)(m+n-1)$

pick red then white = $\frac{m \times n}{(m \times n)}$

pick red then white = $m \cdot n$
a red then any white

pick white then red = $n \cdot m$
a white then any red.

any $R+W$ = $2mn$

$$\therefore P() = \frac{2mn}{(m+n)(m+n-1)}$$

$\{RR; RW; WR; WW\}$

$$P(RR) = \frac{m}{m+n} \cdot \frac{m-1}{m+n-1}$$

$$P(WW) = \frac{n}{m+n} \cdot \frac{n-1}{m+n-1}$$

$$P(WR) = \frac{n}{m+n} \cdot \frac{m}{m+n-1}$$

$$P(RW) = \frac{m}{m+n} \cdot \frac{n}{m+n-1}$$

$$P(\{WR; RW\}) = P(WR) + P(RW)$$

51 b) Roll 3-sided die labelled 1, 2, 3

If k comes up $\rightarrow$ remove k balls from urn

~~P(1R)~~ $\{1R ; 2R ; 3R\}$

~~P(1R)~~ $P(1R) = \frac{1}{3} \cdot \frac{m}{n+m}$

$P(2R) = \frac{1}{3} \cdot \frac{m}{n+m} \cdot \frac{m-1}{n+m-1}$

$P(3R) = \frac{1}{3} \cdot \frac{m}{n+m} \cdot \frac{m-1}{n+m-1} \cdot \frac{m-2}{n+m-2}$

~~$\left(\frac{1}{3} \cdot \frac{\binom{m}{1}}{\binom{n+m}{1}} \cdot \frac{\binom{m}{2}}{\binom{n+m}{2}} \right)$~~

$P(\text{All } R) = \frac{1}{3} \left(\frac{m}{n+m} \right) + \frac{1}{3} \left(\frac{m}{n+m} \cdot \frac{m-1}{n+m-1} \right) + \frac{1}{3} \left(\frac{m}{n+m} \cdot \frac{m-1}{n+m-1} \cdot \frac{m-2}{n+m-2} \right)$

52) 52 card deck. dealt $52!$
 $P(13\text{th card} \rightarrow 1\text{st king } \frac{1}{13}) = ?$

$\rightarrow$ no king in card 1 $\rightarrow$ 12

$\rightarrow$ any of 4 kings.

$\frac{48}{52} \cdot \frac{47}{52}$

~~$\frac{48!}{(52-4)!}$~~

$\frac{\binom{48!}{48!}}{(52-4)!}$

~~(48)~~

~~$\frac{48!}{12!} \left[\frac{48!}{(48-12)!} \right] \left(\frac{4}{4} \right) (4)$~~

~~$\frac{\binom{48!}{36!}}{32!} (4)$~~

~~$P(12) =$~~

$P(\text{no king first 12 cards}) =$

~~$\frac{\binom{48!}{36!}}{(52!)} \left(\frac{4}{40!} \right)$~~

$\frac{\binom{48}{12}}{\binom{52}{12}}$

$P(13\text{th king}) =$

$\frac{4 \binom{48}{12}}{\binom{52}{13}}$

$\rightarrow$ 13th card is a king:

$P(\text{king dealt}) =$

No king 12 = $\binom{48}{12}$

kings = 4 (13th)

total outcomes = $\binom{52}{13}$

prob $K = \frac{4}{52} = \frac{1}{13}$

$P = \left(\frac{1}{13} \right) \cdot \frac{4 \binom{48}{12}}{\binom{52}{13}}$

52) Re-attempt

52 card deck (well shuffled $\rightarrow$ random order)

$$P(13\text{th card is first King dealt}) = ?$$

Notes

- $\rightarrow$ First 12 cards have no king
- $\rightarrow$ 13th card is a king
- $\rightarrow$ 4 distinct kings in 52 card deck.
- $\rightarrow$ draw 13 cards

$$P(\text{King}) = \frac{4}{52} = \frac{1}{13} \quad \times$$

$$\text{total outcomes} = \binom{52}{13}$$

$\hookrightarrow$ Order not relevant
 $\hookrightarrow$ Combinations

$$12 \text{ cards no king: } \binom{48}{12}$$

$\hookrightarrow 48 \cdot 47 \cdot 46 \cdot \dots \cdot 1$
 $\rightarrow 12 \text{ times}$

$$\text{Events} = \frac{(4) \binom{48}{12}}{\binom{52}{13}} = \frac{\binom{48}{12} (4)}{\binom{52}{13}}$$

$$P(13\text{th king drawn}) = \frac{1}{13}$$

$$P(13\text{th card and is first king}) = \frac{1}{13} \cdot \frac{\binom{48}{12}}{\binom{52}{13}}$$

$$P = \frac{1}{13} \times \frac{4 \binom{48}{12}}{\binom{52}{13}}$$

king is last
(given 1 in 13 cards)

prob exactly 1 king
in first 13.

$\triangleright \{ \text{---} \dots \text{---} \}$
13 cards

what is prob king is last slot

$\hookrightarrow$ only 1 event where King
is in slot 13.

$$\hookrightarrow \frac{1}{13}$$

Slots

53) 90 students $\rightarrow$ split 3 classes equal size. $\rightarrow$ 30 per class
Joe / Jane

$$P(\text{Joe + Jane same class}) = ?$$

$$\text{total outcomes} = \frac{90!}{3!(30-3)!}$$

same class:

in one class $\rightarrow$

$$\frac{90!}{30! 30! 30!}$$

$$\frac{88!}{28! 30! 30!}$$

$$P(\text{JJ one class}) = \frac{3 \left(\frac{88!}{28! 30! 30!} \right)}{3 \left(\frac{90!}{30! 30! 30!} \right)} =$$

$$\frac{(3)(88! 28! 30!)(90!)}{3! 30! 30!}$$

$$\left. \begin{array}{l} \text{total outcomes} \\ \text{(for 1 class)} \end{array} \right\} \binom{90}{30}$$

$$\left. \begin{array}{l} \text{JJ in one class} \\ \text{(for 1 class)} \end{array} \right\} \binom{88}{28}$$

$$P(\text{JJ one class}) = \frac{\binom{88}{28}}{\binom{90}{30}}$$

$$P(\text{JJ same of 3 classes}) = 3 \cdot \frac{\binom{88}{28}}{\binom{90}{30}}$$

54) Two car parties (distinct) 20 cars $\rightarrow$ 1 parking lot
10 cars US $\rightarrow$ 20 lots
10 cars foreign

a) ~~20!~~ Different parking config:

$$\text{distinct cars} = 20! \checkmark$$

$$\text{make dependent} = \frac{20!}{(20-2)!} = 380$$

$$b) \# \text{ Alternating Parking Configurations} = \frac{20!}{(20-2)!}$$

$$P = \frac{20 \cdot 19}{20!}$$

$$P = \frac{2 \cdot 10!}{20!} \quad \times$$

$\rightarrow$ Close

$$P = \frac{2 \cdot 10! \cdot 10!}{20!}$$

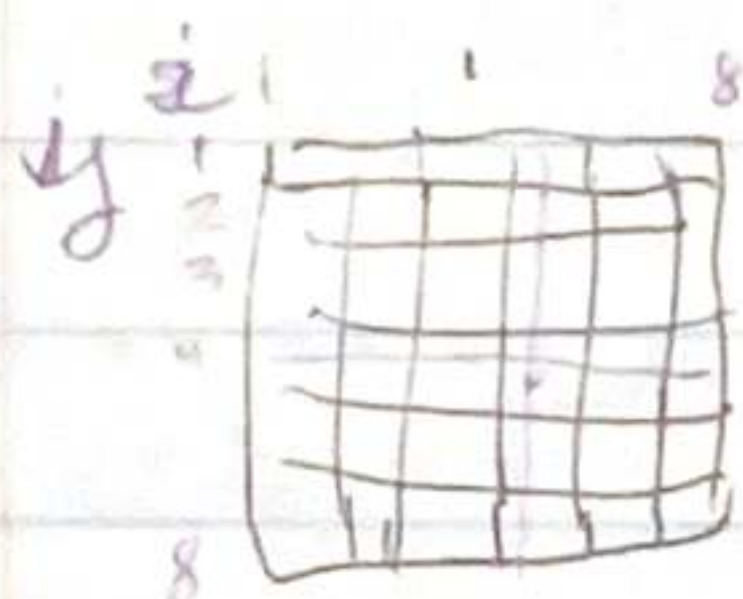
$\rightarrow$ 10! $\rightarrow$ US configs
 $\rightarrow$ 10! $\rightarrow$ F configs

55) 8 Rooks 8x8 Board. (all placements equally likely)

$$P(\text{All Rooks Safe}) = ?$$

$$\text{total placements} = \frac{64!}{(64-8)!} = \frac{64!}{56!}$$

$$\text{total safe rook configs} = 8! \cdot 8!$$



Rook $\rightarrow (3, 1) \rightarrow$ all $(x=3; y=?)$
 $\rightarrow$ all $(x=?; y=1)$

$(x; y)$
 $(1; 3)$
 $(5; 6)$
 $(8; 1)$
 $(7; 3)$

8!

$(x; y)$
 $(1; 8!)$
 $(2; 7!)$
 $(3; 6!)$
 $(4; 5!)$
 $(5; 4!)$
 $(6; 3!)$
 $(7; 2!)$
 $(8; 1!)$

$8! \times 8! \neq$

$$P(\text{All Rooks Safe}) = \frac{8! \cdot 8!}{\frac{64!}{56!}}$$

$$(8! \cdot 8!)$$

$$\frac{64!}{56!}$$

$= \frac{64!}{56!} \rightarrow$ lose Row + Column (7x7 board).

$$\rightarrow \text{Placement matters} \Rightarrow \frac{64!}{(64-8)!} = \frac{64!}{56!}$$

$$P(\text{All safe}) = \frac{8! \cdot 8!}{\frac{64!}{56!}} =$$

56) 8 lower level courses $\{L_1, \dots, L_8\}$
 10 higher level courses $\{H_1, \dots, H_{10}\}$

Curriculum: $4 \times L \quad 3 \times H$

$$a) = \frac{8!}{4!} \times \frac{10!}{3!} \quad \text{X} \Rightarrow \binom{8}{4} \binom{10}{3}$$

b) $\{H_1, \dots, H_5\} \rightarrow \text{req } L_1$
 $\{H_6, \dots, H_{10}\} \rightarrow \text{req } L_2 \& L_3$

$$= \frac{7!}{4!} \times \frac{5!}{3!} + \frac{6!}{4!} + \frac{5!}{3!} \Rightarrow \binom{7}{4} \binom{5}{3} + \binom{6}{4} \binom{5}{3} \quad \text{X}$$

- ① Choose L_1 : ~~$\binom{5}{3} \binom{5}{3}$~~
 ② Choose $L_1 + L_2$ OR L_3 : $2 \cdot \binom{5}{2} \binom{5}{3}$
 ③ Choose $L_1 + L_2 + L_3$: $\binom{5}{1} \binom{10}{3}$
 ④ Choose $L_2 + L_3$: $\binom{5}{2} \binom{5}{3}$

$$\hookrightarrow \binom{5}{3} \binom{5}{3} + 2 \binom{5}{2} \binom{5}{3} + \binom{5}{1} \binom{10}{3}$$

57) 6-word sentences? 26 letters, word $\rightarrow$ nonempty letter sequence
 each letter once

$$\frac{26}{6} \Rightarrow \frac{24}{6} = 4 \rightarrow 2 \quad \text{word} \rightarrow \{1 \text{ letter to } 26 \text{ letters}\}$$

$$\frac{26!}{4!} \binom{26}{6} \quad \text{X} \Rightarrow \binom{26}{6}$$

~~$26!$~~ $26! \rightarrow$ Ways to build a word

$$26! \binom{26-1}{5}$$

$$26! \binom{25}{5}$$



Problem Set - Section 2.2 - Probability Mass Functions

1) 2 games scheduled for one weekend.

$$P(\text{not lose 1st game}) = 0.4 \quad P(\text{Lose \#1}) = 0.3 \quad P(\text{Tie}) = 0.3$$

$$P(\text{not lose 2nd game}) = 0.7 \quad P(\text{Lose \#2}) = 0.15 \quad P(\text{Tie \#2}) = 0.15$$

Win $\Rightarrow$ 2 pts Tie $\Rightarrow$ 1 pt Loss $\Rightarrow$ 0 pt

PMF of points earned:

#	Game	Game State
1		Loss
1		Tie
1		Win

$$P(\text{Win \#1}) + P(\text{Tie \#1}) = 0.4$$

$$P(\text{Lose \#1}) = 0.6$$

$$P(\text{Win \#2}) + P(\text{Tie \#2}) = 0.7$$

$$P(\text{Lose \#2}) = 0.3$$

Let X = number points scored

$$P(X=0) = 0.6 \times 0.3 = 0.18$$

$$P(X=1) = (0.2)(0.3) + (0.3)(0.35) = 0.27$$

$2(\text{Win} + \text{Lose}) + 2(\text{Tie})$

$$P(X=2) = (0.2)(0.35) + (0.2)(0.3) + (0.6)(0.3) = 0.31$$

$$P(X=3) = (0.2)(0.35) + (0.2)(0.3) = 0.14$$

$$P(X=4) = (0.2)(0.35) = 0.07$$

$$P(X \geq 4) = 0$$

2) Party $\rightarrow$ 500 guests
 $P(\text{exactly one other guess} = \text{birthday}) = ?$

365 day 500 guests

$$\text{total} = \frac{500!}{365!} \quad \text{same birthday} = \frac{498!}{364!}$$

$$P(\text{same birthday}) = \frac{\left(\frac{498!}{364!}\right)}{\left(\frac{500!}{365!}\right)} = X$$

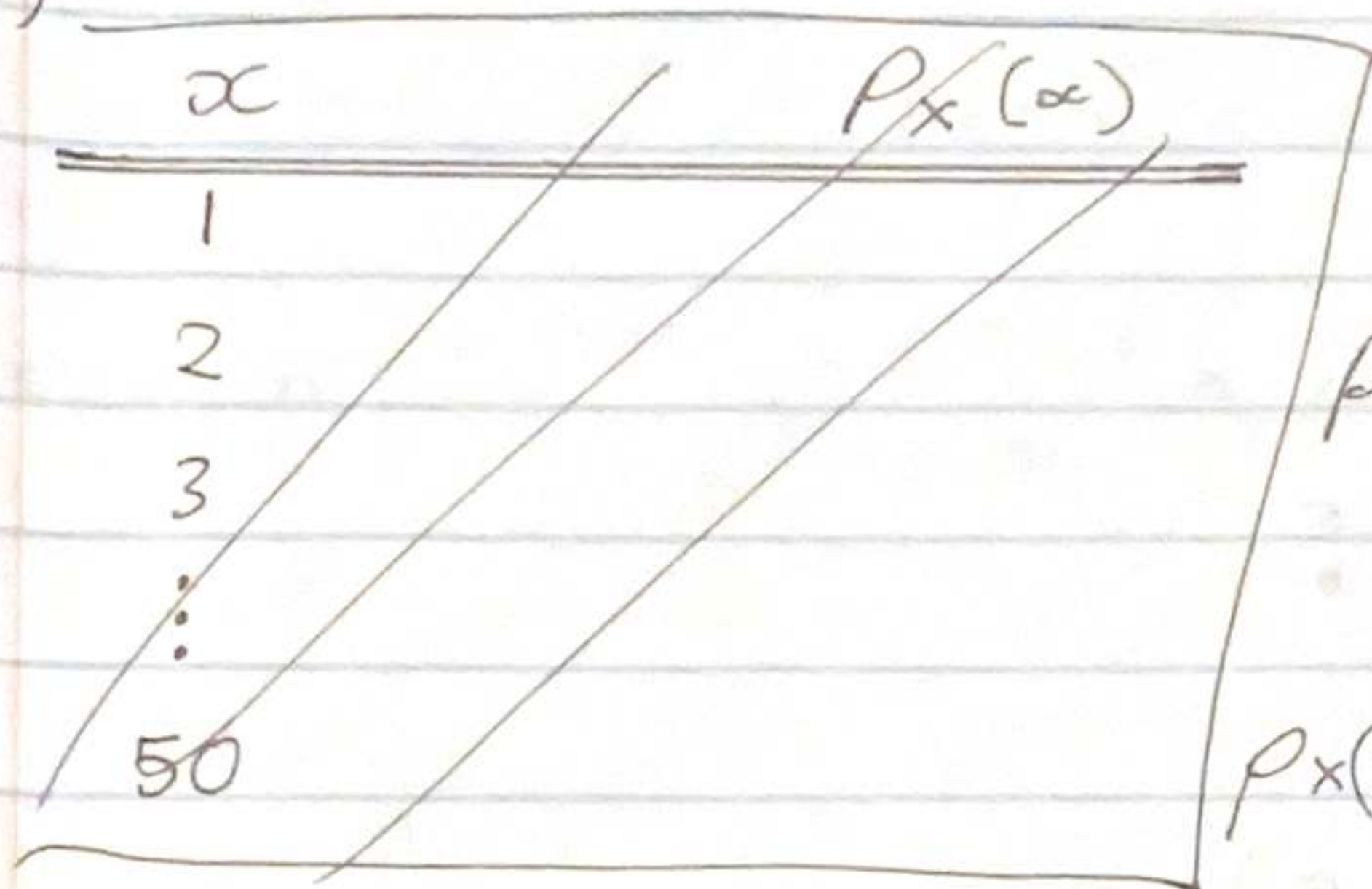
Poisson: $P_x(k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!} \quad (\lambda = \frac{500}{365} ?)$
 $P_x(2) = e^{-2} \cdot \frac{\left(\frac{500}{365}\right)^2}{2!}$
 $= 0,12697999 \quad X$

$\hookrightarrow$ $\binom{499}{1} \times \left(\frac{1}{365}\right) \times \left(\frac{364}{365}\right)^{498}$
 $\hookrightarrow$ 1 other person $\hookrightarrow$ one birthday

$\rightarrow$ # Guest same birthday is binomial.
 $\hookrightarrow$ with $p = \frac{1}{365}$ & $n = 499$

4) ISP $\rightarrow$ 50 modems serve 1000 customers
 $\rightarrow p = 0.01$ % prob each cust need connection at given time.

a).



Binomial MF $\rightarrow$ # successes

$$P_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad (k=0, \dots, n)$$

$$P_X(k) = \binom{1000}{k} (0.01)^k (0.99)^{1000-k};$$

$$k = 0, 1, \dots, 45$$

$$P_X(50) = \sum_{k=50}^{1000} \binom{1000}{k} (0.01)^k$$

b) Poisson: $P_X(k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}$, $k = 0, 1, 2, \dots$

assume $\lambda = 1000(0.01) = 10$ customers

$$P_X(k) = e^{-10} \cdot \frac{10^k}{k!}, \quad k = 0, 1, 2, \dots, 49.$$

$$P_X(50) = \sum_{k=50}^{1000} e^{-10} \cdot \frac{10^k}{k!}, \quad k = 50, \dots, 1000$$

$$P_X(k) = \begin{cases} e^{-10} \cdot \frac{10^k}{k!}, & k = 0, \dots, 49 \\ \sum_{k=50}^{1000} e^{-10} \cdot \frac{10^k}{k!}, & k = 50, \dots, 1000 \end{cases}$$

c) 1) % Bin: $P_X(50) = \sum_{k=50}^{1000} \binom{1000}{k} (0.01)^k (0.99)^{1000-k}$

2) % Poisson (Approx): $P_X(50) = \sum_{k=50}^{1000} e^{-10} \cdot \frac{10^k}{k!}, \quad k = 50, \dots, 1000$

Prob %

1, ..., 49

50, ..., 1000

as modems
scale

insufficient
modems to

scale

6a) n basketball games
 $\hookrightarrow n$ is odd.
 $p = P(\text{Celtics Win})$

a) values of p : $(n=5) > (n=3)$

Binomial: $P_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$; $k=1, 2, \dots, n$

$$P_X(5) = \binom{5}{k} p^k (1-p)^{5-k}$$

$$P_X(3) = \binom{3}{k} p^k (1-p)^{3-k}$$

$$\begin{aligned} \binom{5}{k} p^k (1-p)^{5-k} &\geq \binom{3}{k} p^k (1-p)^{3-k} \\ \binom{5}{k} (1-p)^{5-k} &\geq \binom{3}{k} (1-p)^{3-k} \\ \rightarrow \frac{5!}{k!(5-k)!} (1-p)^{5-k} &\geq \frac{3!}{k!(3-k)!} (1-p)^{3-k} \\ 5! (3-k)! (1-p)^{5-k} &\geq 3! (5-k)! (1-p)^{3-k} \end{aligned}$$

b) $A \rightarrow$ Celtics win with $n = 2k+1$ games

$B \rightarrow$ Celtics win with $n = 2k-1$ games

$N \rightarrow$ # Celtic wins in first $2k-1$ games

Want $P(A) \geq P(B)$, thus $P(A) - P(B) > 0$

$\hookrightarrow P(A) \approx N \rightarrow$ anchor to single reference

$\hookrightarrow$ ask question: What do additional 2 games for $2k+1$ add?

$$P(B) = P(N \geq k) \quad \text{oo win with games above } k.$$

6 b) con.) A $\rightarrow$ Celtics win with $n = 2k+1$ games
 B $\rightarrow$ Celtics win w/ $n = 2k-1$ games

N $\rightarrow$ # wins in first $2k-1$ game
 $\hookrightarrow$ anchor / reference point

wins required % for $2k-1$ % $\frac{(2k-1)+1}{2} = (k)$ wins
 for $2k+1$ % $\frac{(2k+1)+1}{2} = (k+1)$ wins

$$P(B) = P(N \geq k) \quad \# \text{ win condition reached}$$

$P(A) =$ multiple cases

case 1 % $= P(N \geq k+1)$ # win condition reached

case 2 % $= P(N = k) \cdot P(\text{at least 1 win in 2 games})$ # one win short
 $= P(N = k)P(1 - (1-p)^2)$

case 3 % $= P(N = k-1) p^2$

case 4 % $N \leq k-2$ # too far behind to win in 2 games.

$$\therefore P(A) = P(N \geq k+1) + P(N = k)(1 - (1-p)^2) + P(N = k-1)p^2$$

$$\begin{aligned} \Rightarrow P(A) - P(B) &= P(N \geq k+1) + P(N = k)(1 - (1-p)^2) + P(N = k-1)p^2 - P(N \geq k) \\ &= P(N = k) \cdot (1 - (1-p)^2) - P(N = k-1)p^2 + 1 \\ &= P(N = k)(1 - (1-p)^2 - 1) + P(N = k-1)p^2 \\ &= \cancel{P(N = k)(1 - (1-p)^2 - 1)} \\ &= P(N = k-1)p^2 - P(N = k)(1-p^2) \end{aligned}$$

8) $X \rightarrow$ binomial random variable (n, p)

$$P_X(k+1) = \frac{p}{1-p} \cdot \frac{n-k}{k+1} \cdot P_X(k)$$

$$P_X(k+1) = \binom{n}{k+1} p^{k+1} (1-p)^{n-k-1}$$

$$P_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\frac{P_X(k+1)}{P_X(k)} = \frac{\binom{n}{k+1} p^{k+1} (1-p)^{n-k-1}}{\binom{n}{k} p^k (1-p)^{n-k}}$$

$$= \frac{\binom{n}{k+1} p^{k+1} (1-p)^{n-k-1}}{\binom{n}{k} p^k (1-p)^{n-k}}$$

$$= \frac{\binom{n}{k+1} p (1-p)^{-1}}{\binom{n}{k}} = \frac{p}{1-p} \cdot \frac{\binom{n}{k+1}}{\binom{n}{k}}$$

$$= \frac{p}{1-p} \cdot \frac{n!}{(k+1)!(n-k-1)!} \cdot \frac{k!(n-k)!}{n!} = \frac{p}{1-p} \cdot \frac{k!(n-k)!}{(k+1)!(n-k-1)!} \cdot \frac{k!(n-k)!}{k!}$$

$$= \frac{p}{1-p} \cdot \frac{n-k}{k+1} \cdot \left(\frac{1}{(k+1)} \right) \cdot \left(\frac{n-k}{1} \right)$$

$$= \frac{p}{1-p} \cdot \frac{n-k}{k+1}$$

10) ~~1x matchbox~~ ~~1x matchbox~~ ~~1x matchbox~~ ~~1x matchbox~~

11) 1x matchbox L pocket
1x matchbox

• Section 2.3 Problem Sets - Functions of RVs

- 13) ~~5 natural mix~~ 5 natural mix $P(B/N) = 0.5$
 2 adopted girls $P(G/N) = 0.5$

$$P_N(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{+10}$$

$$\begin{aligned} P_N(k) &= \binom{5}{k} p^k (1-p)^{5-k} \quad \text{+10} \\ &= \binom{5}{k} (0.5)^k (0.5)^{5-k} \quad \text{+10} \\ &= \binom{5}{k} \left(\frac{1}{2}\right)^5 \end{aligned}$$

$G = \#$ girls out of 7 children.

$$G = N + 2$$

$$P_G(g) = \sum_{\{n: n=g+2\}} P_N(n) = P_N(g-2)$$

$$\therefore P_G(g) = \begin{cases} \binom{5}{g-2} \cdot \left(\frac{1}{2}\right)^5 & ; \text{ if } 2 \leq g \leq 7 \\ 0 & ; \text{ otherwise} \end{cases}$$

- 14) ~~$X \rightarrow$ RV $0 \dots 9$ equal prob: $p = 1/10$~~

a) ~~$Y = X \bmod(3)$~~

~~$$\begin{aligned} P_X(k) &= \binom{n}{k} p^k (1-p)^{n-k} \\ &= \binom{10}{k} (0.1)^k (0.9)^{10-k} \end{aligned}$$~~

~~$$P_Y(y) = \sum_{\{x: X \bmod(3)=y\}} P_X(x) = P_X\left(\frac{y}{\bmod(3)}\right)$$~~

~~$$P_Y(y) = \binom{10}{\frac{y}{\bmod(3)}} (0.1)^{\frac{y}{\bmod(3)}} (0.9)^{10 - \left(\frac{y}{\bmod(3)}\right)} \quad ; \text{ if } 0 \leq y \leq 2$$~~

X

see next page

14) Solution Workthrough
 $0 \dots 9 \pmod{3}$

possible values = $\{0; 1; 2\}$

a)

$$P_Y(0) = P_X(0) + P_X(3) + P_X(6) + P_X(9) = \frac{4}{10}$$

$$P_Y(1) = P_X(1) + P_X(4) + P_X(7) = \frac{3}{10}$$

$$P_Y(2) = P_X(2) + P_X(5) + P_X(8) = \frac{3}{10}$$

$$P_Y(y) = 0 \text{ if } y \notin \{0, 1, 2\}.$$

$$\frac{4}{10} + \frac{3}{10} + \frac{3}{10} = 1 \quad \text{check.}$$

b) $Y = 5 \pmod{(X+1)}$

x	0	1	2	3	4	5	6	7	8	9
$P_X(x)$	0	3	2	1	0	5	5	5	5	5

$$P_Y(0) = P_X(0) + P_X(4) = \frac{2}{10}$$

$$P_Y(1) = P_X(3) = \frac{1}{10}$$

$$P_Y(2) = P_X(2) = \frac{1}{10}$$

$$P_Y(3) = P_X(1) = \frac{3}{10}$$

$$P_Y(5) = P_X(5) + P_X(6) + P_X(7) + P_X(8) + P_X(9) = \frac{5}{10}$$

$$P_Y(y) = 0 \text{ if } y \notin \{0, 1, 2, 3, 5\}$$

$$\frac{2}{10} + \frac{1}{10} + \frac{1}{10} + \frac{3}{10} + \frac{5}{10} = 1 \quad \text{check}$$

$$P_Y(y) = \sum_{\{x: 5 \pmod{(x+1)} = y\}} P_X(x)$$

15) K ; equal prob $P = \frac{1}{2n+1}$ $[-n; n]$

$Y = \ln X$; $X = a^{|K|}$; $a > 0$ $\ln(a) = x$
 $a = e^x$
 $\ln a^{|K|} = |K| \ln a$

$P_X(y) = \sum_{\{x: \ln x = y\}} P_X(x)$

$Y = K \ln a$

$$P_Y(y) = \begin{cases} \frac{2}{2n+1} & \text{if } y = \ln a, 2\ln a, \dots, n\ln a \\ \frac{1}{2n+1} & \text{if } y = 0 \\ 0 & \text{otherwise} \end{cases}$$

• Problem Set 2.4 - Expectation, Mean, Variance

16) a)

$$a \neq 0; a \in \mathbb{R}$$

$$E[X] = \sum_x x \cdot p_x(x)$$

$$= \sum_x x \cdot \frac{x^2}{a}$$

$$= \frac{1}{a} \sum_x x \cdot x^2$$

$$\sum_x p_x(x) = 1 = \sum_x \frac{x^2}{a}$$

$$a = \sum_{x=-3}^3 x^2$$

$$a = (-3)^2 + (-2)^2 + (-1)^2 + 0 + (1)^2 + (2)^2 + (3)^2$$

$$= 9 + 4 + 1 + 0 + 1 + 4 + 9$$

$$= 18 + 8 + 2$$

$$= 28$$

$$a = 28$$

$$E[X] = \sum_x x \cdot p_x(x) = \sum_{x=-3}^3 x \cdot \frac{x^2}{28}$$

$$= (-3) \frac{(-3)^2}{28} + (-2) \frac{(-2)^2}{28} + (-1) \frac{(-1)^2}{28} + (0) + (1) \frac{(1)^2}{28} + (2) \frac{(2)^2}{28} + (3) \frac{(3)^2}{28}$$

$$= -\frac{9}{7} + 0 + \frac{9}{7} = 0$$

$$E[X] = 0$$

18) $a, b \rightarrow$ pos ints. ; $a \leq b$; equal prob $[2^a; 2^b]$

$$p_x(x) = \begin{cases} \frac{1}{b-a+1} & ; \text{ if } x = 2^k, a \leq k \leq b, \text{ integer} \\ 0 & ; \text{ otherwise} \end{cases}$$

$$E[X] = \sum_x x \cdot p_x(x)$$

$$= \sum_{k=a}^b (2^k) \cdot \left(\frac{1}{b-a+1} \right)$$

$$2^{m+1} - 1 = 1 + 2 + 2^2 + \dots + 2^m$$

$$m = b - a$$

$$= \left(\frac{1}{b-a+1} \right) [2^a + 2^{a+1} + \dots + 2^{b-a}]$$

$$= \frac{2^a}{b-a+1} (1 + 2 + \dots + 2^{b-a})$$

$$= \frac{(2^a)(2^{b-a+1} - 1)}{b-a+1}$$

$$= \frac{2^{b+1} - 2^a}{b-a+1}$$

Geometric Series

$$\sum_{k=0}^m r^k = \frac{r^{m+1} - 1}{r - 1}$$

$$\begin{aligned}
 E[X^2] &= \sum_{x=a}^b x^2 \cdot p_x(x^2) \\
 &= \sum_{k=0}^b 2^{2k} \left(\frac{1}{b-a+1} \right) \\
 &= \left(\frac{1}{b-a+1} \right) \left(2^{2a} + 2^{2(a+1)} + 2^{2(a+2)} + \dots + 2^{2(b-a)} \right) \\
 &= \left(\frac{2^{2a}}{b-a+1} \right) \left(1 + 2 + \dots + 2^{b-a} \right) \quad \frac{2^n - 1}{2 - 1} = \dots \\
 &= \frac{(2^{2a})(2^{b-a+1} - 1)}{b-a+1} \quad \frac{(4^a)(4^{b-a+1} - 4^a)}{b-a+1} \\
 &= \frac{2^{b+a+1} - 4^a}{b-a+1} \quad \frac{2^{b(a+1)} - 4^a}{b-a+1} \quad \frac{4^{b+1} - 4^a}{3(b-a+1)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X) &= E[X^2] - (E[X])^2 \\
 &= \frac{4^{b+1} - 4^a}{3(b-a+1)} - \left(\frac{2^{b+1} - 2^a}{b-a+1} \right)^2
 \end{aligned}$$

20) Find ticket = p

$$p_x(x) = (1-p)^{x-1} \cdot p, \quad k=1, 2, 3, \dots$$

$$E[X] = \frac{1}{p}$$

$$\text{var}(X) = \frac{1-p}{p^2}$$

$$= E[X^2] - (E[X])^2$$

$$= (1-p)^{x^2-1} \cdot p - [(1-p)^{2x-2} \cdot p^2]^x$$

$$= p(1-p)^{-1} [(1-p)^{x^2} - p(1-p)^{2x-1}]$$

$$= p(1-p)^{-1} [(1-p)^{x^2} - p(1-p)^{2x} \cdot (1-p)^{-1}]$$

22) 2x coins tossed until coin : 1 → heads
2 → tails

coin 1 → heads : p
coin 2 → ~~heads~~ tails : q

success = pq ✗

PMF: $P_X(k) = (1 - pq)^{k-1} (pq)$ ✗ ; $k = 1, 2, \dots$

$E[X] = \frac{1}{pq}$ ✗

$\text{var}(X) = \frac{1 - pq}{(pq)^2}$ ✗

$P(\text{HT} ; \text{TH}) = p(1-q) + q(1-p)$ ✓

Geometric PMF: $P_X(k) = [1 - p(1-q) - q(1-p)]^{k-1} (p(1-q) + q(1-p))$

$E[X] = \frac{1}{p(1-q) + q(1-p)} = \frac{1}{p - pq + q - pq} = \frac{1}{p + q - pq^2}$ ✓

$\text{var}(X) = \frac{1 - [p(1-q) + q(1-p)]}{[p(1-q) + q(1-p)]^2} = \frac{1 - p + pq - q + pq}{p^2(1-q)^2 + 2pq(1-p)(1-q) + q^2(1-p)^2}$

$= \frac{2pq - p - q + 1}{p^2 - p^2q^2 + 2q^2(1 - 2p + p^2) + q^2(1 - 2p + p^2)}$

$= \frac{1 - p(1-q) - q(1-p)}{[p(1-q) + q(1-p)]^2}$ ✓

$P_X(k) = \left\{ \begin{aligned} &= \frac{1 - p + pq - q + pq}{\dots} \\ &= \frac{1 - p - q + 2pq}{\dots} \\ &= \frac{pq + (pq + 1 - p - q)}{\dots} \\ &= \frac{pq + (1 - p)(1 - q)}{\dots} \end{aligned} \right.$

Specific form
but answer equivalent,

• Problem Set 2.5 - Joint PMFs + Multiple RVs

- 24) 100 shares stock A $X \text{ \& } Y \rightarrow \text{price changes A \& B}$
 200 shares stock B PMF is uniform over set of integers $x \text{ \& } y$
 $\hookrightarrow -2 \leq x \leq 4,$
 $-1 \leq y - x \leq 1,$
 $\hookrightarrow (x-1 \leq y \leq x+1)$

a) $P_{X,Y}(x,y) =$
 $x = \{-2; -1; 0; 1; 2; 3; 4\}$
 $y = \{-3; -2; -1; 0; 1; 2; 3; 4; 5\}$

$$R = \{(x,y) \mid -2 \leq x \leq 4, -1 \leq y - x \leq 1\}$$

$$\# \text{ pairs} = 3 \cdot 7 = 21$$

is uniform distribution $\therefore p = \frac{1}{21}$

$$P_{X,Y}(x,y) = \begin{cases} \frac{1}{21} & ; \text{if } (x,y) \text{ is in } R \\ 0 & ; \text{otherwise.} \end{cases}$$

Marginals:

$$P_X(x) = \sum_y P_{X,Y}(x,y)$$

$$= P_{X,Y=-3}(x,-3) + P_{X,Y=-2}(x,-2) + P_{X,Y=-1}(x,-1)$$

$$= \frac{1}{21} + \frac{1}{21} + \frac{1}{21} = \frac{3}{21}$$

$$P_X(x) = \begin{cases} \frac{3}{21} & ; \text{if } x = -2, -1, \dots, 4 \\ 0 & ; \text{otherwise} \end{cases}$$

$$E[X] = 1$$

$$P_Y(y) = \sum_x P_{X,Y}(x,y)$$

$$P_Y(y) = \begin{cases} 7/21 & ; \text{ if } y = \{-3, -2, -1\} \\ 0 & ; \text{ otherwise} \end{cases} \quad \times$$

$E[Y] =$

x	Y
-2	-3, -2, -1
-1	-2, -1, 0
0	-1, 0, 1, 2
1	1, 2, 3
2	2, 3, 4
3	3, 4, 5
4	3, 4, 5

$$P_Y(y) = \begin{cases} 1/21 & ; \text{ if } y = -3 \\ 2/21 & ; \text{ if } y = -2 \\ 3/21 & ; \text{ if } y = \{-1, 0, 1, 2, 3\} \\ 2/21 & ; \text{ if } y = 4 \\ 1/21 & ; \text{ if } y = 5 \\ 0 & ; \text{ otherwise} \end{cases}$$

$$E[Y] = \frac{1}{21}(-3 + 5) + \frac{2}{21}(-2 + 4) + \frac{3}{21}(-1 + 0 + 1 + 2 + 3)$$

$$= \frac{2}{21} + \frac{4}{21} + \frac{15}{21}$$

$$E[Y] = \frac{21}{21} = 1$$

b) mean of trader's profit $\rightarrow$ full joint ^{pricing} PMF + weight

$$P = 100X + 200Y$$

$$E[X, Y] = 100E[X] + 200E[Y]$$

$$= 100(1) + 200(1)$$

$$= \underline{300}$$

26) golf score $\{101; 110\}$ $p = 0.1$
 days $\{x_1, x_2, x_3\}$

a) PMF:

~~$P_{X,Y}(x,y) =$~~

~~$P_{X,S}(x,s) =$~~ $P_X(k) = \left\{ \right.$

~~...~~

$$P_X(k) = \begin{cases} P(X > k-1) - P(X > k) & ; k=101, 110 \\ 0 & ; \text{otherwise} \end{cases}$$

$$P(X > 100) = 1$$

$$\begin{aligned} \therefore P(X > k) &= P(X_1 > k, X_2 > k, X_3 > k) \\ &= P(X_1 > k)P(X_2 > k)P(X_3 > k) \\ &= \left(\frac{100-k}{10} \right)^3 \\ &= \frac{(100-k)^3}{10^3} \end{aligned}$$

$$\therefore P_X(k) = \begin{cases} \frac{(110-(k-1))^3}{10^3} - \frac{(110-k)^3}{10^3} & \end{cases}$$

$$P_X(k) = \begin{cases} \frac{(111-k)^3 - (110-k)^3}{10^3} & ; \text{if } \dots \\ 0 & ; \text{otherwise} \end{cases}$$

b) $E[X] = 105.5$ $\frac{101+110}{2} = 105.5$

$$E[X] = \sum_{k=101}^{110} x \cdot \frac{(111-k)^3 - (110-k)^3}{10^3} = 103.025$$

$$\text{Imprament} = 105.5 - 103.025 = 2.475$$